Land Records Modernisation in Odisha: A Study of History, Policy and Implications

Brief of the Research

This thesis aims to undertake a critical analysis of the transformation of the presumptive nature of land records and ownership to a conclusive land title regime in India. Also, it aims to examine the impact of the concerted policy push under the Digital India Land Records Modernization Programme (DILRMP) on the condition of land records and cadastral maps, the process of registration and mutation, and the diverse land tenure system in the State of Odisha, India. The study intends to sift through various theoretical discourses such as right to property as an institution, land as a commodity, and rights in land as an instrument to expand freedom. It assesses the modernisation of land records, maps the existing record of rights (RoR) to the possession, understands the workings of the revenue system and its interaction with people, and reviews the use of RoR to avail farmers' welfare schemes in the State. It proposes to explore the discrepancies between spatial and textual data, as well as ownership and possession of land in the location of the study (two villages in Nayagarh district, Odisha). As a result, it also expects to reflect on landlessness, tenancy and sharecropping, decline of commons, and land parcelisation and concentration among the rural population.

Well-defined property rights are the cornerstone of the liberal argument for economic growth. In this paradigm, the state is there to protect property rights and enforce contracts, which facilitates innovation and growth. Thus, interests in formalising rights in land emerge, and accurate land records become crucial for such needs. Here the formalisation process followed by the state includes official registration and issuance of titles, which brings a crucial distinction between possession of an asset and ownership of an asset.

So, in section I, we give a brief overview of theoretical discourses surrounding property rights, their benefits, and the nature and processes of formalisation of the rights in land.

In regards to property rights in land, India is a unique case because land records in India continue to have presumptive status, which means the entries in the record of rights are presumed to be correct unless the contrary is proved. However, India seeks to transform the existing presumptive nature of rights in land into the conclusive title to land. For the same, the government has been modernising records management, integrating registration with

mutation, and digitising land records (both spatial and textual). Digital India Land Records Modernization Programme (DILRMP), starting in 2016, has expedited the process of modernisation in many States in India. The State of Odisha is considered one of the leading States that have shown remarkable progress in the implementation of DILRMP.

Therefore, in section II, we explore the nature of property rights in India and Odisha in particular, the status of implementation of DILRMP in the State and its challenges. Also, we try to make a case for research by providing a methodology for the same.

I

Literature Review

Secure private property, as a concept, has been the mainstay of the liberal argument for economic growth. It is argued that economic growth is natural and logical in the regime of secure private property. While discussing the importance of property rights, we must remember that the system of exclusive, transferable, alienable, and enforceable private property is a result of massive economic, social, and political changes (North, 1981). So, the preconditions available in the West cannot be presumed in the developing countries that have not witnessed such changes. Although the concept is not alien to developing countries, the wholesale import for analysis is akin to building the roof without the foundations. Therefore, we analyse the concept from various schools of thought for a holistic understanding.

1.1 Right to Property in the Neoliberal Discourse

The neoliberal discourse has roots in neoclassical economic theory. The system of private property emerged long before the advent of neoclassical theory. However, this branch of economic thought systematises and rationalises the importance of private property according to its own central assumptions. When we take those assumptions as given, private property becomes inevitable.

The neoclassical theory has three main assumptions. First, people have rational preferences among outcomes. Second, individuals maximise utility and firms maximise profits. And, third, people act independently on the basis of full and relevant information.

Besides these central assumptions, the neoclassical model assumes zero transaction costs and a lack of asymmetric information. Transaction cost is the cost agents bear while participating in the market. It is the sum of the cost of bargaining, transferring, gathering information, and involving the state as an enforcer between the parties entering into a contract. Asymmetric information refers to the imperfection and asymmetry with respect to the availability of information with the contracting parties.

New Institutional Economics (NIE), an extension of neoclassical theory, works with the assumptions that transaction costs and asymmetric information exist in the real world, even as it furthers the analysis based on utility maximisation and rational preference. New institutional theories provide a paradigm for the emergence of property rights in different contexts. For instance, when land is abundant, the gains received from the assignment of any property rights are very low. As the land becomes scarce, agents are incentivised to get into an enhanced arrangement of property rights. With resources getting scarce, the communal arrangement to property rights emerges to keep the outsiders at bay. The enforcement of community rights happens to be effective when a community is small and the penetration of technology is not wide. And when new market opportunities arise, because of trade and entrepreneurship, individual property rights give more incentives to the agents.

When transaction costs of a property are low, the initial assignment of property rights is not a matter of concern, because two individuals can trade between themselves to come to an arrangement to increase production. With the increase in transaction costs, agents do not voluntarily come together to promote growth. They evade formal agreements, which results in property rights not being enforced by formal institutions. Information is fairly symmetric among the members of a small community. Members of the community have similar information about ownership, possession, and rights of transfer accorded to its members, but an increase in population and migration gives rise to information asymmetry. In a more advanced stage of development, it is no longer possible to keep all the transactions among the community members, and information asymmetry increases. Therefore, both transaction cost and asymmetric information have to be minimised for the efficient functioning of the property rights institutions in the new institutional framework. According to Hodgson (1993), the new institutional approach contrasts with the seminal work of Karl Marx and Max Weber, who explained the origin and development of a culture of rational maximisers as a specific historical phenomenon, rather than as a universal feature of human life.

In the neoliberal framework, Harold Demsetz's contribution to understanding social benefits and costs with respect to the emergence of property rights has been insightful. The Demsetz (1967) thesis argues that a property regime develops when its social benefits exceed its social costs. According to his thesis, no kind of property regime is specific to and a preserve of any particular community or ethnic identity. The native Canadians, in the Labrador Peninsula, developed private property rights because of the advent of the fur trade in the 17th century. However, Native Americans in the southwestern plains of the United States did not establish private property rights, since the animals they hunted had wide tracts of land to wander about and it was not feasible to establish private hunting lands there.

Demsetz argues that there are three distinct benefits of property rights. First, they produce incentives for the development of resources. Second, they reduce rent dissipation, which means assets get devalued because of unrestricted access. Third, private property reduces the number of parties who have to come to a consensus to install measures to control negative externalities.

According to Alchian & Demsetz (1973), the owner of a property owns a bundle of socially recognised rights to use the resources, and not the resource itself. While it recognises one person's interests against others, ownership of a party can be shared by the use rights of another party. Legal theorists contemporarily advocate this concept of the bundle of rights as it opens the pathways for the state to intervene — a departure from the neoliberal framework.

Moreover, well-defined property rights in their operation provide benefits with respect to (a) the creation of incentives with tenure security, (b) the improvement of transferability, and (c) the ability to use land as collateral to access formal credit (Deininger & Feder, 1998). These benefits, of course, vary in quantum and proportion in different contexts, but it is important to understand the respective benefits.

Tenure security is about the temporal nature of security in rights. If the rights are secured for the long term, assured by a strong enforcement mechanism, it eliminates the future risk of loss of income. This creates an opportunity for the owners to make productivity-enhancing long-term investments (Besley, 1995). Transferability is an important factor in the land market. The ability to transfer land from one user to another depends on the availability of information and trust. In a small community, in the absence of information asymmetry, it

might be seamless. But when the community becomes more integrated geographically, secure property rights facilitate transactions between individuals from different communities.

The system of property rights in land has an impact on the credit market. Creditors as they give out loans have to reduce uncertainty and moral hazard problems. So, the mechanism of collateral arrangements emerges, where fixed assets, most often land, are used as a guarantee for loan repayment (Feder & Nishio, 1998). Land has traditionally been a good collateral to secure loans, and land with a title is more so. However, lands with titles do not necessarily create a credit market, which is a more complex institution based on savings and income.

Feder et al. (1988) conducted an important econometric study that showed that agricultural land in Thailand without a formal title was worth only one-half to two-thirds of land with a formal title. Further, increasing the security of private property rights also led to an increased value of the crops produced. More secure titling also provided better access to credit and led to greater investments (Feder & Feeny, 1991).

Hernando de Soto advances this specific argument that a formal title has consequential benefits and gives a grand theory on development and capitalism. In his book *'The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else'*, de Soto (2000) suggests that lands under the possession of the poor in non-western countries do not have proper titles, and as there is no proper system of formalisation and enforcement, the assets owned by the poor can be understood as *'dead capital'*. And when the problem of the dead capital is solved, capitalism takes root and resolves the problem of economic backwardness. However, there is a fundamental problem in his analysis. He considers that lack of proper rights is the source of the problem in developing countries. Rather, the lack of formal rights is a manifestation of a bigger problem. Taking Douglass North's argument that history matters and the system of exclusive, transferable, alienable, and enforceable private property is a product of history, this inversion of logic by de Soto that fixing one institution paves the way for capitalism is misplaced.

Empirically, it is not established that the legalisation and formalisation of land occupied by slum dwellers and squatters has led to growth in the formal credit market (de Olarte, 2001). According to Woodruff (2001), except for a few exceptions, none of such efforts of turning extra-legal assets into legal ones has succeeded. There are no instances where assets were transformed into capital. Formalisation of an asset is important, but regarding it as capital is

misplaced. Also, the view that the poor own capital, nonetheless dead in nature, pre-empts any redistributive measures.

1.2 Right to Property from the Agrarian Perspective

Demsetz's thesis and the legalist view recognise property as a bundle of rights, i.e., the owner of a property owns a bundle of socially recognised rights to use the resources. This view of social relations, however, does not capture the hierarchy and power in social relations. So, we must broaden the horizon of understanding by incorporating the agrarian relations present on the ground. Ownership or possession of land is reflective of the social relationship and the material conditions existing in an agrarian economy and life (Ramachandran & Ramakumar, 2000)**.** The ownership of most of the land, as well as of the best, belongs to the landlord households in Indian villages, where the social relations are divided between and recognised as tenants, hired workers, and the owner (Ramachandran, 2011).

The ownership of land with respect to the structure of classes, along with their identities such as caste, gender and others, guides the agrarian perspective on land rights. So, any study on property rights in India must address the issues and concerns with regard to the land rights of landowners, tenants, sharecroppers, women, and land concerns of the landless.

The agrarian perspective furthers the instrumental view of freedom, wherein formalised property rights in land help advance the general capability of a person. According to Sen (1999, p.10), from the instrumental perspective, there are five distinct types of freedom: political freedoms, economic facilities, social opportunities, transparency guarantees, and protective security. Among all of them, transparency guarantees have a clear instrumental role in preventing corruption and underhand dealings. Formalised property rights clear the concealed and ill-defined nature of ownership and pre-empt corruption and rent-seeking, while shedding light on agrarian relations and land concentration, which is of great value in a democracy. It provides transparency not just for an exchange in the marketplace, but also for democratic engagement among citizens.

Moreover, the agrarian perspective provides a universal value to formalisation. It recognises that formalisation has an intrinsic value and an instrumental use. If de Soto's advocacy of formalisation is about the realisation of the *'dead capital'*, from the agrarian perspective it is about creating an institution for it has an intrinsic value. The neoliberal motivation behind de

Soto's argument is to create capital out of assets, which helps the poor to participate in the market, whereas the instrumental perspective shines a light on the amount and nature of assets in possession by the poor vis-à-vis the rich.

The neoliberal motivation also pervades the policy prescriptions for developing countries. In fact, all policy proposals by the World Bank insist on the importance of private property because it creates growth in the land market. For Deininger & Feder (1998), "Land registration and titling systems are often perceived as an important element in a policy seeking to promote tenure security and to facilitate more effective markets". The World Bank view prioritises land as a commodity over all other values: property rights over various social processes and social relations. Thus, land becomes an opportunity for investment and speculation. In the market economy, every production, distribution and sale is controlled by the market alone. This applies to goods, but, according to Polanyi (1957), "labour, land and money are essential elements of industry", but cannot be produced themselves; thus, they cannot be seen as commodities. Labour is another name for a human activity that "goes with life itself", and cannot be produced for gain. Land is another name for nature, which is not produced at all. Besides this, the neoliberal state uses land as a hook to attract domestic and foreign capital. This marks the policy shifts in land records management in India from state-led land reform to the policy of facilitating the growth of the land market (Nayak, 2013). For instance, an initiative in the State of Karnataka, named *Bhoomi*, started in 2000, which aimed to computerise land records and provide online delivery of revenue documents, aided big businesses to locate in the rapidly urbanising peripheries of Karnataka's capital city, Bengaluru (Benjamin & Raman, 2011). In the process, the traditional claims over land were sidelined to reconfigure the force fields of land in favour of large developers. Such arguments undermine the conceptualisation of land as a bearer of rights and an instrument to gain freedom. The idea of secure private property also becomes somewhat tenuous from this commodification perspective.

Moreover, the property as a single right defined in terms of alienation and exclusion does not help us understand the existence of the commons and the use of natural resources by the community (Schlager & Ostrom, 1992). But, increasingly, the idea of alienation and exclusion has become inescapable because of privatisation recommended by public policies that cause depletion of common property resources (CPRs). According to Jodha (1989), the depletion of common property resources happens mainly for four reasons: demographic

factors, market factors, technological factors, and public interventions. Public interventions including institutional reforms such as formalisation of rights accentuate the role of other factors in depleting CPRs.

1.3 Nature and Processes of Formalisation of Land Rights and Titling

When we think of formalisation of land rights, the concept of 'titling' becomes important. The process of recording of transfer of rights by an official is called registration, which entails the issuance of titles. Both together make the formalisation process complete.

Formalisation brings a crucial distinction between possession of an asset and ownership of an asset. The continued presence of an individual on a parcel of land is evidence that the land belongs to the person or he has possession. However, the possession or the use of the land does not necessarily prove the ownership of the person. Ownership is a social fact, an agreed-upon concept by the community, individual and the state (Bromley, 2008). When the state is completely legitimised through its governance and authority, the title bestowed by it proves the ownership. Otherwise, a title can only be called a low-cost legal instrument that lacks the force it claims to have. A title is a promissory note that promises the full backing of the government to protect the titleholder's interest in the thing that the title is an abstraction of. A titleholder holds the right to call upon the state to protect his rights.

Deed Registration and Title Registration

The formalisation of property rights, namely, through the instrument of a title, constitutes primarily two legal documents that provide a title to land: deed registration and title registration. Under the deed registration system, the transfer is registered as a deed by a centralised public office that gathers and organises all registrations. In the title registration system, the state guarantees and records the transfer. In the deed registration, a buyer has to verify the history of transactions in the land and ascertain himself the current ownership, and when satisfied, he executes a deed with the owner. In the title registration, all the documents relating to a land are submitted to the title registration officer, and after verification, he allows for the registration of the title to happen. Both legal documents are produced through different processes while registering the same interests. And the only difference is that in the title registration, the state guarantees the nature of transfer.

The title registration system gives evidence of the current ownership, providing full-poof updated ownership status backed by the state. However, deed registration, even after helping mutation in the record of rights in the state document, provides a presumptive ownership status in many countries.

The title registration system has three main principles as it provides a title. First, the mirror principle, which means the title mirrors the ground reality. Second, the curtain principle, which represents the idea that the title brings closure to all disputes to the date. Third, the indemnity principle, which guarantees insurance in the form of compensation in case of any error.

This example will illustrate the difference between both systems and bring clarity to their operation. For instance, 'A' has the ownership of a piece of land, which is sold to 'B' by a third person 'C' fraudulently. If the title is produced under the title registration system, the person who bought the land, 'B', is protected. And if the title is generated under the deed registration system, the original owner 'A's' rights are restored, and the buyer 'B' loses the land. In the first case, 'C' can be sued, and 'A' is entitled to insurance.

In the registration of title, a person who has the title surrenders his title, so that a fresh title is generated for the new titleholder. So, how does a person get his title registered in the first place? The person has to apply for the same when the institution of title registration is established in a country. This is the operational mechanism: If a landholder wants to get his title registered, he approaches the title registration office with all his documents to prove his ownership. The title registration officer checks the history of ownership and issues notices to all who might have any interest in the said land. He checks the encumbrances and liens against the property. After the search operation, the officer starts or stays the proceedings depending on how the case is made out. In case of the titleholder's demise, the title does not automatically go to the heirs, but they have to go through the process again to register the title.

In deed registration, parties involved in a contract generate a sale deed approved by the state. Here, the state is involved twice and works in silos; later, when conflict arises, the court has the authority to decide upon the ownership. This system became unpopular and was deemed inefficient at the advent of colonial expansion in the countries where land was abundant, namely, Australia. Therefore, the system of title registration was conceived, so that the new

colonial administration started granting titles. This drew curtains on various legacy issues as well as the aboriginal history to start the rights in land afresh, by the authority of the King. The system was named after Sir Robert Torrens — 'Torrens System' — and was first introduced in South Australia in 1858.

Niblack (1915) highlights the contentious points in the Torrens system. The system accords an important power to the registrar of land titles, as he provides the indefeasible and conclusive title. The act of title registration becomes a judicial act, rather than an administrative and clerical act. The registrar performs a judicial function, as the generated title is judicial in nature and gives a verdict on ownership. This point is contentious as to how the office can perform the role of the judiciary. At the same time, it takes away some accountability from the registrar: he decides to confer the title based on the documents available before him, but when an error crops up, the person who is deprived of his rights is compensated by an indemnity fund.

Torrens system poses serious difficulties in its implementation. It requires drastic changes in the existing system for the presence of multiple records handled by state departments that work in silos. There is also a possibility of a valid title owner being deprived of his property because of corruption, forgery, and the inability of the registrar to unveil layers of interests at the time of registering the title. The aggrieved party has the only option to get compensation from the insurance fund. Private insurance companies might not cover all the defects and risks. When a true legal owner loses his rights because of the inefficiency and corruption in the system, it is a violation of fundamental principles of natural justice. So, even as the Torrens system is institutionalised and accepted in some Western countries, the system of appeal against the errors coexists.

II

2.1 Property Rights in India and the Need for Formalisation

The rights in land in India evolved historically and, later, were deeply shaped by the British. The rights were more or less similar throughout Indian history until the British came, who configured the land rights, in a modern and capitalist method, to accumulate capital and collect taxes. According to Appu (1996), traditionally all Indian rulers, both Hindu and Muslim, appropriated a share of the produce from the land for the state revenue. However,

during the decline of the Mughal Empire, with the slipping away of control on the collection of revenue, the provincial governors resorted to appointing feudal chieftains, government functionaries, and moneylenders – the powerful class who had little connection with the land – as revenue farmers. The British administrators treated them as landlords and entered into agreements that they collect and pay a sum of money (fixed or variable) in exchange for the status quo.

Rothermund (1971, p. 1) argues that the British didn't intervene in the land revenue system of India, but they made important changes in the system of revenue settlements by bringing in concepts such as rent, wages, cost of production, and private property rights. The property rights system in India was based on 'interests' attached to or on the land. But the British administration favoured exclusive claims of ownership to ensure a steady flow of land revenue.

The Permanent Settlement of 1793 laid the foundation of the Indian legal system by the British. The judiciary was set up as an independent organ of the state to protect property rights. However, the protection was only based on the appropriation of land revenue, rather than the security of title. When the rights were recorded, it was meant to fix the responsibility of the people for the payment of revenue. As long as revenue was collected on a land, it was acceptable for the administration and the judiciary. No attempt was made to look into the actual ownership of the land. The records of land were revenue records, not of the titles (ibid, p. 2).

As the institutions did not provide the holder with a secure title, it was pragmatic on the part of the British to give the land records a presumptive status. The record of rights (RoR) continues to be a presumptive proof of ownership. Courts may disregard the record if they have proof that is more conclusive.

The report of the *Task Force on Agrarian Relations by Planning Commission* in 1973, as quoted by (Wadhwa, 2002), mentions "the absence of up-to-date land records is a serious obstacle in the implementation of land reforms". Several other reports have also highlighted that the main reason for the non-implementation of land reforms is the lack of reliable records of rights. The importance of the updation of records was never realised until the late 1980s. It was Manmohan Singh, the then deputy chairman of the Planning Commission, who appointed a one-man committee in 1987 to study the problem of record of rights in land in

the country. The committee under D. C. Wadhwa recommended that the present system of presumptive titles to land be converted to conclusive titles to land.

During the Seventh Five-Year Plan (1985-1990), two ambitious schemes called Computerisation of Land Records (CLR) and Strengthening of Revenue Administration and Updating of Land Records (SRA & ULR) were launched by the central government. In 2008, both the schemes were merged and the National Land Records Modernization Programme (NLRMP) was launched. This was a big step to move from the present system to the Torrens system. The Guidelines for NLRMP mentions (Government of India, 2008-09):

> The programme would modernise management of land records, minimise scope of land/property disputes, enhance transparency in the land records maintenance system and facilitate moving eventually towards guaranteed conclusive titles to immovable properties in the country.

Then, in 2016, the NLRMP programme was renamed as the Digital India Land Records Modernization Programme (DILRMP) with a broader mandate and scope. It aimed to achieve the following outcomes in land records management with the implementation of the DILRMP:

> 1) Integration of spatial databases (cadastral map) with textual Record of Rights (RoR).
>
> 2) (a) Delivery of citizen services like digitalised maps and computerised RoR (b) Online issuance of digitalised maps and computerised Record of Rights, (c) Online transmission of required information from the integrated land information management system from one office to another.
>
> 3) Provide online single window at-a-glance access to all available and relevant information to give a fair comprehensive position of any plot of land in question (Government of India, 2018-19).

There is an emphasis on an integrated information management system in DILRMP. It aims to integrate the departments that collect and produce different land records, and work in silos. The revenue department in a State maintains textual records (RoR). The survey and settlement department manages the Field Management Book (FMB) and village/cadastral maps. The department of stamps and registration maintains sale deed. Historically, all three departments have worked in silos. Records collected and generated by one department do not necessarily reflect on the records of another department. For instance, a record executed by

the registration department, namely, sale deed, does not automatically reflect in the records of revenue department, i.e., RoR, without mutation. Meanwhile, the cadastral maps managed by the survey and settlement department only change with resurveys, which is not a recurrent activity and has not happened in many villages since the 1960s.

2.2 History of Revenue System in Odisha

Odisha has had a history of heterogenous system of landownership and tenancy. In 1936, when Odisha was formed, it inherited different land revenue systems from different parts of the British administration. The Northen and the coastal Odisha — regions such as Cuttack, Puri, Balasore, Anugul, etc. — continued to have *zamindari* system, the dominant system under the Bengal Presidency. The southern Odisha, regions such as Ganjam and Koraput, got its peasant-proprietor system of *ryotwari* from the Madras Presidency. The western regions such as Sambalpur, Nuapada, etc. inherited their system from the Central Provinces. Apart from this, the *gadjat* areas, which constituted 24 former princely states, had special arrangements with the British.

Under these four main land tenure types, several divergent interests in land evolved among and between landlords and *raiyats* (Jena, 1957). Various land tenure holders were recorded by the British for revenue collection: rent-free tenure holders, holders paying quit rent, holders paying a fixed nominal rent (*zamindars*), and holders paying full rent (*raiyats*). Also, village-based revenue collection was prevalent through village heads such as *sarbarakars*, *gountias*, and *pradhans*.

Gaurav & Mishra (2016), quoting Padhi (1999), state that the regional diversity in the development of Odisha has roots in the land relations and the revenue systems of precolonial and colonial times. After Independence, such diversity in land revenue system and the inequities embedded in them presented a reason for the State to enact various land reform legislations. Despite the legislations, implementation has been less than a success and access to land remains a distant dream for a vast majority of the people. Moreover, according to Sinha (2014), from his action research project in the Puri district of Odisha, there is a reversal of earlier successes, as revenue officials connive with landowners and delete entries in the record of rights, which dispossess the assignees who have been cultivating the ceiling surplus land.

2.3 Computerisation of Land Records in Odisha and DILRMP

Computerisation of land records was first introduced in the State of Odisha in 1988-89 on a pilot basis against the backdrop of CLR and SRA & ULR. The modernisation started amidst the advent of neoliberalism, the discourse of 'Guaranteeing Title to Land', and the Wadhwa Committee report in 1989. However, it was not actively operated until the early 2000s. The full-scale implementation of the project was started from 2003-04. The National Land Records Management Programme (NLRMP) of 2008, the program formed after merging CLR and SRA & ULR, gave an impetus to the computerisation of land records in Odisha. And DILRMP starting in 2016 brought a new dynamism to the modernisation of land records.

Singh & Behera (2012) found that by 2012 despite the fact that there was an improvement in the efficiency in land records management and service delivery in Odisha, the awareness among public about computersation of land records was low, and many landholders still depended on intermediaries for any kind of land records issues, which is a clear case of rent-seeking.

DILRMP is aimed at developing a modern comprehensive and transparent land records management system. The scope of the programme mainly includes: (a) computerisation of land records, (b) survey/resurvey and updating of the Survey & Settlement Records, (c) computerisation of registration, and (d) modern record rooms/land records management centres at tehsil level. Following these larger goals, all textual data of RoRs have been updated and computerised, regularly reviewing the status of disposal of pending mutation cases. Digitisation of all cadastral maps has been completed. *Bhulekh* (for textual) and *Bhunaksha* (for spatial) websites are regularly updated. High-tech survey operation is being undertaken across districts. To replace the existing system of physical storing and maintenance of land records, the government has created Modern Record Rooms in all 317 tehsils of the State, where the land records are indexed and stored electronically. All tehsils and revenue offices are provided with the State Wide Area Network (SWAN) and are now connected through OSWAN. The integration of the land records with registration, through web-based mutation application software to automate mutation following registration, is complete.

Table 2: Progress of DILRMP in Odisha vis-a-vis India

	India	Odisha
Computerisation of Land Records (RoRs)	6, 22,653/6,57,417 (94.71% revenue villages)	51,700/51,700 (99.95% revenue villages)
Automation of Sub-Registrar Offices	4924/5303 (92.85% SROs completed)	191/191 (100% SROs)
Integration of Land Records and Property Registration	4035/5303 (76.09% completed)	191/191 (100% completed)
Survey/Resurvey	91,459/6,57,417 (13.91% of villages completed)	3130/51,700 (6.04 % completed)
Digitisation of Cadastral Maps	75.64% completed	100% completed
Bhunaksha Customisation	62.06% of villages completed	100% completed
Textual and Spatial Data Integration	67.48% of villages completed	100% completed
Tehsil Modern Record Room	3324/6931 (47.96% of tehsils completed)	308/317 (97.16% of tehsils completed)

Source: Gathered from the DILRMP website (Department of Land Resources, 2023)

2.4 Integrated Process of Registration and Mutation in Odisha

Typically, mutation cases constitute more than 60 per cent of total revenue cases in a tehsil. And district sub-registrar office mostly deals with cases of land registration. So, it is imperative to understand how registration and mutation happen after the modernisation.

First, the registering officer verifies and ensures that OLR Form No. 3 is filled with all relevant information and affixed with proper court fees. After doing the needful, he scans and uploads Form No. 3 on E-Registration system. The officer generates a transmission report (for Form No. 3) from the MIS report of E-Registration system. Also, he forwards the hard

copy of Form No. 3 to the concerned tehsildar. The concerned tehsildar downloads the form and the verification process begins at the tehsil office.

Second, after receiving Form No. 3, tehsildar institutes the case and forwards it to the concerned RI. Tehsildar has to start the case maximum within 10 days. If the tehsildar does not institute the case within that window, the case will be auto-instituted and forwarded to RI (after reporting to sub-collector). The RI has to dispose of the case within 30 working days at least after one hearing. In case the RI does not conduct the hearing within time and the case is pending beyond 30 days, it will be auto-escalated to the tehsildar who ensures that the work is done. Once the mutation is confirmed, the work of the record keeper starts. *Khata* no and *plot* no. get auto-generated. Record keeper views the case, adds rent and cess, and updates details on his software. Then he forwards it to the office in charge for approval. Meanwhile, map correction is made both by *Amin* and RI. RI/*Amin* does the measurement by visiting the site. Then, RoR is generated and dispatched to the concerned address via speed post, and after that, RoR is updated on the *Bhulekh* website.

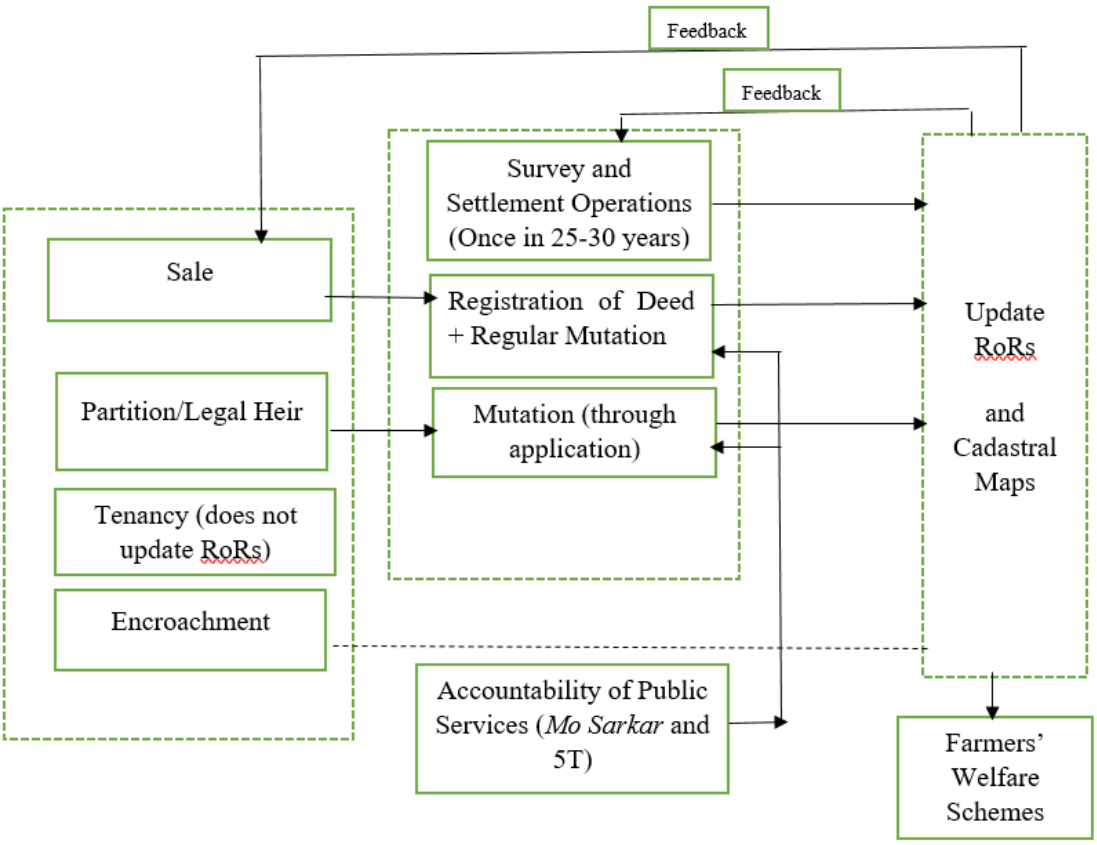
As per this process, within 90 days, an uncontested mutation case is disposed of and the concerned RoR is dispatched to the citizen. In 2020, the Odisha government amended the Odisha Survey and Settlement Rules (1962) as well as the mutation manual. According to the new rules, uncontested mutation cases, which represent 70 per cent of the total mutation cases, can be disposed of within a week. An appeal against the same can be presented within 180 days. This appeal period is not a bar for correction of records. This amendment and the reduction in the days for case disposal fit squarely with the Odisha government's 5T and *Mo Sarkar* programmes, bringing dynamism to the delivery of public services.

Challenges in Updation of Record of Rights in Odisha

Even with the integration of registration and mutation, there are many challenges in the updation of RoRs. During land sales and purchases, many transactions go unrecorded, as the process of mutation is still considered lengthy and expensive from the perspective of the public. In the case of partible inheritance, each individual plot is subdivided among many heirs. In most cases, legal heirs do not undertake partition through legal means, and the land continues to be under the name of an ancestor. In Odisha, legislations exist to prevent encroachment on government-owned 'wastelands', and to transfer up to an acre of 'unobjectionable' public land to landless families, but these are largely ineffective, because

there are powerful incentives for the officials to take bribes from encroachers to permit continued cultivation, rather than to initiate eviction proceedings or formalisation. The Orissa Land Reforms Act prohibits sub-letting of land, regulates rents, and grants occupancy rights to long-standing tenants. In spite of such restrictions, tenancy remains widely prevalent. Most of the contracts are oral. Sharecropping is the predominant form of tenancy contract in Odisha. Updation of RoR does not capture the problem of widespread tenancy.

A Framework to Analyse the Interaction Between Institutions and Instruments



This chart will dictate the fieldwork and the analysis to understand the interaction (sale, partition, and encroachment) between private individuals and the state. It will also guide our research with respect to record of rights as an instrument and the farmers' welfare schemes.

2.6 Rationale of the Study

There is a growing consensus among various stakeholders to formalise land rights. However, while building any institution to cater to the demands of the present, we must remember that history matters and the present is connected to the past by the continuity of the nature of land

ownership, tenures, and tenancy. Moreover, the complete resurvey has been done only in six per cent of villages of Odisha, illustrating the fact that only six per cent of spatial data (cadastral map) is updated. Although it is claimed that almost 100 per cent of non-spatial data (land records) is computerised, most of the RoRs have legacy issues. Only what goes through the mutation process is updated and published on the *Bhulekh* website. So, this gap between the RoRs and the conditions on the ground makes a case for study.

In Odisha, two important websites track textual and spatial data: *Bhulekh* is the repository for RoRs (a mix of both old and new); *Bhunaksha* stores the cadastral maps (mostly the old, from the last Survey and Settlement). Apart from this, the revenue department maintains a dashboard that tracks the status of mutation cases and shows the number of cases that are disposed of regularly. However, this mechanism does not capture the number of pending cases that do not enter the mutation process in the first place – umpteen number of partition, legal heir, and encroachment cases do not visit the system.

The Survey and Settlement is the process by which the state reaches the ground and provides an equal opportunity for all the cases to be heard and resolved. However, the Survey and Settlement is a protracted process that requires political will, which is limited in supply as resurveys lay bare unwarranted problems for both bureaucracy and government. Therefore, it is important to ask if all such alternative methods are a substitute for a resurvey. And how far technology can resolve the conflicts existing on the ground? At best, it provides an opportunity for everyone to check and know the unknowable, but in a village, most are aware of the situation and are unwilling to address it for various reasons, which must be explored and documented.

As the shift from presumptive title to conclusive title is extensive, DILRMP pushes for incremental technological changes that include the modernisation of records and maps and integration and coordination among various offices of the revenue department. The stated objective is to eliminate all sorts of ambiguities in the land and dispense conclusive titles. However, the design of the program is suited to and aligned with the neoliberal policies and the welfare programs of the government. While updated RoRs have their benefits in terms of conflict resolution, they facilitate the process of land acquisition for development projects without recognising various traditional claims over land and result in land alienation and encroachment. In addition to aiding development projects, modernisation of land records

plays an important role in the state welfare programs and in targeting the beneficiaries. For instance, a farmer can apply for the farmers' welfare schemes after procuring an updated RoR from the website. For official use, RoR becomes a useful document to check the eligibility of an applicant. To be precise, having an instrument like RoR does not necessarily make someone included in a scheme, but to claim the benefits the instrument becomes necessary. So, it is imperative to understand the role of an updated RoR in the inclusion of a person in the schemes, and his exclusion for the lack of an updated RoR.

As our study aims to investigate the status of records and identify major challenges and unmitigated errors, it is important to understand whether the infusion of technology is making an impact on the said challenges, or remains technocratic.

DILRMP proposes to address the issues of land governance, thus the program must be understood in terms of broad policies of neoliberal reforms. DILRMP should not be studied just as a policy, because it meets society and family (inheritance and legal heir) on the ground at a sociological level. Therefore, we aim to study the interaction between different schools of thought such as land as a commodity, land rights as an instrument, and land as social relations.

2.7 Scope of the Study

The research aims to contribute to the literature on the formalisation of land records in general and to the literature on the workings of DILRMP in the State of Odisha in particular. During the study, we aim to document the workings of the bureaucracy: document institutional rules, operations, and their effects. It plans to capture the whole dynamism of registration and mutation and the role of various people involved in it. It aims to touch high-level bureaucrats, collector, sub-collector, tehsildar, RI, *Amin*, registration officers, people who write sale deeds, the kiosk owners, and the local touts. As far as the scope of the research goes, it aims to provide a template for further studies on the modernisation of land records in India.

2.8 Research Questions

1. What is the progress of land records modernisation in Odisha? What are the objectives of the same and how far are they achieved?
    - Here we try to locate the motivation behind land records modernisation, and how the program is designed and implemented by the Odisha government.
2. Are the objectives and design of the modernisation programme in harmony with the outcomes? Are the updated RoRs used towards different ends by different sections of society?
    - How does a RoR help a farmer to take advantage of welfare schemes? How does a RoR enable someone to buy and sell land? How does the updation of RoR turn out to be useful in case of land acquisition, and in the worst-case scenario, results in land grabbing and decline of CPR?
3. How is the history of land ownership and tenancy related to the current ownership?
    - Here we inquire if the land ownership of the past has an effect on current ownership through either inheritance or sale. Land parcelisation, partition, and disputes surrounding family ownership will be understood.
4. What is the source of disjuncture between the land records and possession/ownership on the ground?
    - Here we assess the discrepancies between spatial and textual data as well as ownership and possession of land.
5. Are the recent efforts towards the modernisation of land records technocratic?
    - Here we assess the computerisation, the process of data input and verification through biometrics and Aadhaar during registration and mutation.
6. How have the alternative methods to the Survey and Settlement been able to bring parity and uniformity among different records?
    - Alternative methods include the integration of registration to mutation, updation of RoR on *Bhulekh* website, updation of cadastral maps on *Bhunaksha* website, and the High-tech surveys. The Unique Land Parcel Identification Number (ULPIN) assigns a 14-digit alphanumeric number to land parcels based on the geo-coordinates.

2.9 Methodology

The study will employ an exploratory method of research using surveys, qualitative and quantitative analysis, and information from other literature and archives. The surveys include land survey (using Gunter's chain) and household survey, wherein both together map land possession and ownership of the family with the records. The analysis of the quantitative data as well as the interviews remains in the realm of observational studies rather than establishing causal inferences among the variables.

Part I: Top-level Assessment

- An assessment of the structural design of the land administration system of Odisha, institutions and their rules, web portals, services, and grievance redressal.
- A review of the existing legislative, regulatory, and judicial framework governing access to land in Odisha to identify specific consequences for the rural poor.
- Discussions with principal stakeholders and officials involved in policy-making and land revenue administration will be conducted.
- A review of the progress of DILRMP at the State-level. The official data released by the State with respect to its progress will be documented and analysed.
- An assessment of a district in Odisha, Nayagarh, will be done with respect to its history and land tenure relationships, as well as the progress of modernisation of records in the district.

Part II: Ground-level assessment

- We identify two villages from the Nayagarh district and do household surveys to understand land concentration, tenancy, and landlessness among the villagers. We select the villages based on the presence of development projects and welfare policies, as well as on a pilot basis to inquire upon the questions we have set out to explore.
- Select 15 sample parcels (with the consent of the landholder) from each village to assess the discrepancies between the actual ownership and possession of the land. For each plot, the RoR available on *Bhulekh* and the cadastral map from *Bhunaksha* will be matched and mapped with the help of physical surveys.
- Interviews with villagers about their experience with the land administration system (about technology, awareness of websites, corruption, and rent-seeking).

- Interviews with villagers about the clarity of their own records and the disputes, if any, with neighbours and/or family members. Interviews explore the context of CPR and the decline thereof.
- Assessment of the usage of RoRs to avail Kalia scheme (Krushak Assistance for Livelihood and Income Augmentation). The inclusion and the exclusion based on the presence of RoRs will be checked.

We believe it is not desirable to attempt to administer a formal, structured questionnaire within a statistically rigorous sampling frame. Rather, semi-structured interviews will be conducted with individual informants. In order to understand the practical functioning of the land administration system, considerable flexibility, as well as rapprochement, is required, which would not be possible with standardised techniques. However, in the State-level and district-level assessments, we have to create a template for documentation and reporting.

Probable Chapters

1. Literature Review: Discourses around Property Rights and Land Titling
2. Revenue Administrative System of Odisha and its workings
3. DILRMP and Odisha (state level) [motivation, policy objectives, and achievements]
4. Modernisation of Land Records in Nayagarh (district level)
5. Village Level Assessment of RoR and Maps
6. Sociology of Land Ownership and Land Transfers vis-à-vis Land Records
7. Conclusion

References

Alchian, A., & Demsetz, H. (1973). The Property Right Paradigm. *The Journal of Economic History, 33*(1), 16-27.

Appu, P. (1996). *Land Reforms in India: A Survey of Policy, Legislation and Implementation.* New Delhi: Vikas Publishing House.

Benjamin, S., & Raman, B. (2011). Illegible Claims, Legal Titles, and the Worlding of Bangalore. *Revue Tiers Monde, 206*, 37-54.

Besley, T. (1995). *Property Rights and Investment Incentives: Theory and Evidence from Ghana.* Journal of Political Economy.

Bromley, D. (2008). Formalising Property Relations in the Developing World: The Wrong Prescription for the Wrong Malady. *Land Use Policy, 26,* 20-27.

de Olarte, E. G. (2001). Hernando de Soto's Mysteries [Review of The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else, by H. de Soto]. *SAIS Review (1989-2003),* 275-282.

de Soto, H. (2000). *The Mystery of Capital: Why Capitalism Triumphs in The West and Fails Everywhere Else.* New York: Basic Books.

Deininger, K., & Feder, G. (1998). *Land Institutions and Land Markets.* Development Research Group, Rural Development. The World Bank.

Demsetz, H. (1967). Toward a Theory of Property Rights. *The American Economic Review, 57*(2), 347-359.

Department of Land Resources. (2023). *Digital India Land Records Modernization Programme (DILRMP-MIS 2.0).* Retrieved from https://dilrmp.gov.in/#: https://dilrmp.gov.in/#

Feder, G., & Feeny, D. (1991). Land Tenure and Property Rights: Theory and Implications for Development Policy. *5*(1), 135–153.

Feder, G., & Nishio, A. (1998). The Benefits of Land Registration and Titling: Economic and Social Perspectives. *Land Use Policy, 15*(1), 25-43.

Feder, G., Onchan, T., Chalamwong, Y., & Hangladoran, C. (1988). Land Policies and Form Productivity in Thailand. *Johns Hopkins University Press*.

Gaurav, S., & Mishra, S. (2016). Land Ownership and Tenancy in Odisha. In P. B. Nayak, S. Panda, & P. K. Pattanaik, *The Economy of Odisha: A Profile.* Oxford.

Government of India. (2008-09). *The National Land Records Modernization Programme (NLRMP): Guidelines, Technical Manuals and MIS.* Department of Land Resources Ministry of Rural Development .

Government of India. (2018-19). *The Digital India Land Records Modernization Programme (DILRMP): Guidelines, Technical Manuals and MIS* . Department of Land Resources Ministry of Rural Development .

Hodgson, G. M. (1993). Institutional Economics: Surveying the 'Old' and the 'New'. *Metroeconomica, 44*(1), 1-28.

Jena, D. (1957). Effectiveness of Provision in the Tenancy Legislation in Orissa for (i) Security of Tenure and (ii) Improvement in the Tenurial Status of Tenants. *Indian Journal of Agricultural Economics, 12(2),* 135-141.

Jodha, N. (1989). Depletion of Common Property Resources in India: Micro-Level Evidence. *Population and Development Review, 15*, 261-283.

Nayak, P. (2013). Policy Shifts in Land Records Management. *Economic & Political Weekly, 48*(24), 71-75.

Niblack, W. C. (1915). *Pivotal Points in the Torrens System* (Vol. 24). The Yale Law Journal.

North, D. (1981). *Structure and Change in Economic History.* New York: Norton.

Padhi, S. (1999). *Land Relations and Agrarian Development in India: A Comparative Historical Study of Regional Variations.* New Delhi: Oxford.

Polanyi, K. (1957). *The Great Transformations: The Political and Economic Origins of Our Time.* Boston: Beacon Press.

Ramachandran, V. K. (2011). Classes and Class Differentiation in India's Countryside. *World Review of Political Economy, 2(4).*

Ramachandran, V. K., & Ramakumar, R. (2000). Agrarian Reforms and Rural Development Policies in India: A Note. *Paper presented at the International Conference on Agrarian Reform and Rural Development, Department of Agrarian Reform, Government of the Philippines and the Philippines Development Academy.*

Rothermund, D. (1971). The Record of Rights in British India. *The Indian Economic and Social History Review, 4*, 443-461.

Schlager, E., & Ostrom, E. (1992). Property Rights Regimes and Natural Resources: A Conceptual Analysis. *Land Economics, 68*(3), 249-262.

Sen, A. (1999). *Development as Freedom.* Oxford University Press, Oxford.

Singh, P., & Behera, H. (2012). *Assessment of Computerization of Land Records in Orissa.* LBS National Academy of Administration.

Sinha, B. (2014). Land Reforms: Evidence of Reversal in Orissa. *Journal of Land and Rural Studies*, 171-190.

Wadhwa, D. (2002). Guaranteeing Title to Land. *Economic and Political Weekly, 37*(47), 4699-4722.

Woodruff, C. (2001). Review of de Soto's "The Mystery of Capital" [Review of The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else, by H. de Soto]. *Journal of Economic Literature, 39(4)*, 1215–1223.